9C Lesson 2

Immunity and vaccination

Year 9 Science · 9C Threat from disease · Student Book pages 39 to 41

Name _____ Class _____ Date _____

Today I can:

☐ I can describe how phagocytes and lymphocytes destroy microorganisms in the body.

☐ I can explain why you only get many infectious diseases once, and how vaccination protects people and communities.

Key words

white blood cell a blood cell that fights pathogens

phagocyte a white blood cell that surrounds, ingests and digests microorganisms

lymphocyte a white blood cell that makes antibodies

antibody a protein that sticks to one type of pathogen

specific fits only one type of pathogen

memory cell a lymphocyte that stays in the body ready to make antibodies quickly

immune protected against a disease, so you do not become ill with it

vaccine a substance containing parts of a pathogen, given to make a person immune

virus a very small pathogen that can only copy itself inside a cell

dose the amount of a medicine given at one time

half-life the time for the amount of a medicine in the body to fall by half

● Core Everyone does this section.

1. Complete the text using words from the word bank. Some words are used more than once.

When you are infected by a _____, white blood cells start to attack.

_____ ingest some of the microorganisms, while _____ start to produce

_____ that stick to the outside of the microorganisms. These proteins may cause the

_____ to burst open, or cause several microorganisms to _____ together,

which makes it easier for them to be _____.

2. Match each key word to its meaning. Draw a line.

phagocyte	a lymphocyte that stays in the body, ready to make antibodies quickly
lymphocyte	a protein that sticks to one type of pathogen
antibody	a white blood cell that ingests and digests microorganisms
memory cell	contains parts of a pathogen and makes a person immune
vaccine	a white blood cell that makes antibodies

3. Tick one box in each row to show whether the statement is true or false.

Statement	True	False
You are ill before you get better because new antibodies take days to make.		
One type of antibody fits every type of pathogen.		
A vaccine gives you a mild case of the disease.		
Memory cells make the second response faster and larger.		
Vaccinating most people can stop a disease spreading in a community.		

●● Stretch When Core is finished.

S1. A child who has had measles is immune to measles but not to chickenpox. Explain why.

S2. Explain how vaccination against measles protects a child from becoming ill with measles.

S3. Viruses cause diseases such as chickenpox and polio, but they are not classed as living organisms. Suggest why.

●●● Challenge Explain in full sentences.

C1. A vet uses an antibiotic called ceftiofur. The table gives the dose and the half-life for some animals. The half-life is the time for the amount of antibiotic in the animal's body to fall by half.

Species	Dose (mg of antibiotic per kg of animal)	Half-life (hours)
alpaca	6.6	45
American black duck	6.6	32
ball python	10	64
goat	15	37

a) State what is meant by the dose of a medicine.

b) To be effective, the concentration of ceftiofur in a ball python must stay above 2.5 mg/kg. Calculate how many days after the first dose a second dose is needed. Show your working.

c) Calculate the dose, in mg, for a 15 kg goat.

C2. Explain how vaccinating most of the people in a community protects people who cannot be vaccinated, such as newborn babies.

My thinking (any language)

A flu vaccine has to be given every year, but one course of polio vaccine protects for life. What might be different about the flu virus?

Word bank pathogen · phagocytes · lymphocytes · antibodies · stick · ingested · memory cell · immune · vaccine · specific

✂ ---------- cut here

Exit ticket 9C Lesson 2: Immunity and vaccination Name _____

1. Name the two types of white blood cell and say what each one does.

2. Why do you only get measles once?

3. What does a vaccine contain, and why does it not give you the disease?

Today I can: Goal 1 ☐ Goal 2 ☐